

KiCad (7.0) for the University of Glasgow PCB Manufacturing Services

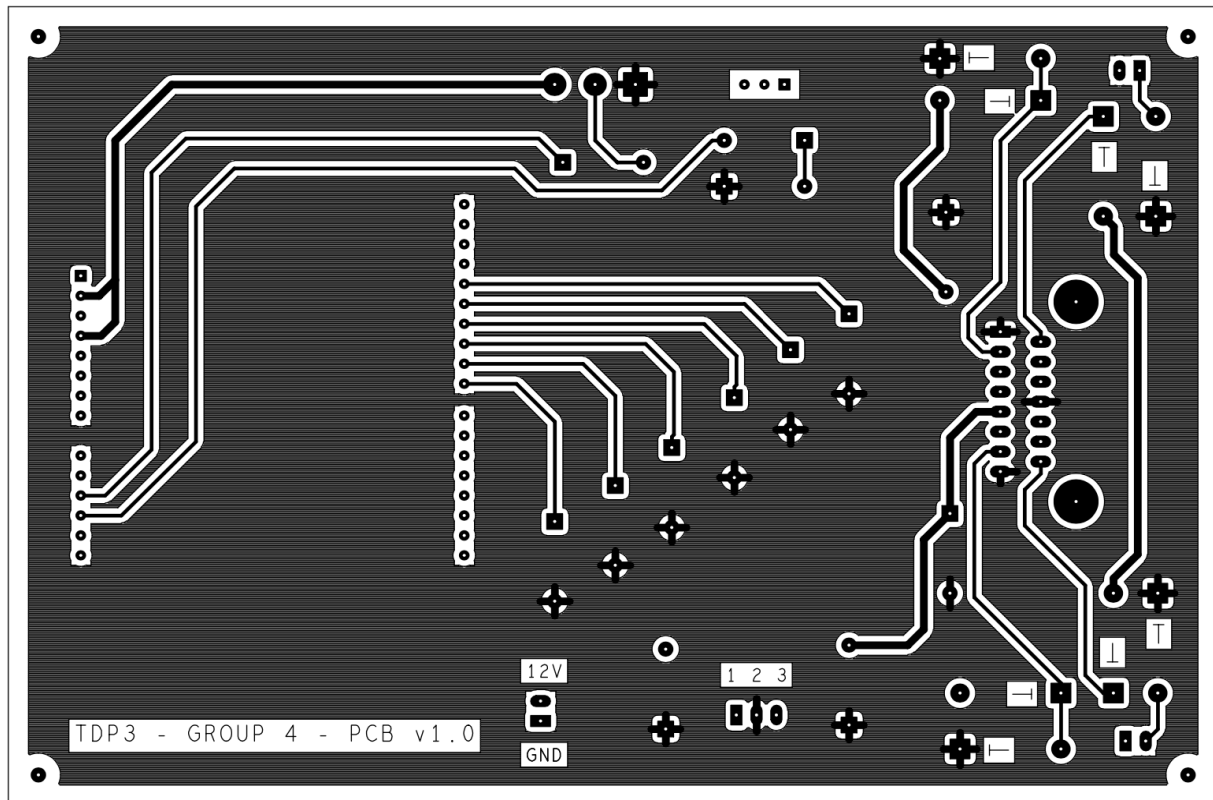
Written by Nicolas Viale -

v1.1 - 13/02/2024

(Update suggestions from Andrew Phillips, and new section 4.2)

1 - Introduction and limitations

This document is based on the OrCad PCB guide given by the university¹ to produce a board compatible with the PCB manufacturing they offer. The manufacturing process is based on photomasks and a job request can be sent as an OrCad board file, or with photomasks and drill files. The aim is to provide help on how to get those production files from KiCad 7.0 and how to set the PCB constraints to match manufacturing capabilities.



- PCB photomask example with top view.

This guide does not teach how to use KiCad nor how to properly design a PCB, it is assumed that you know how to design a schematic, assign footprints, import the design to the PCB editor and create a PCB in KiCad.

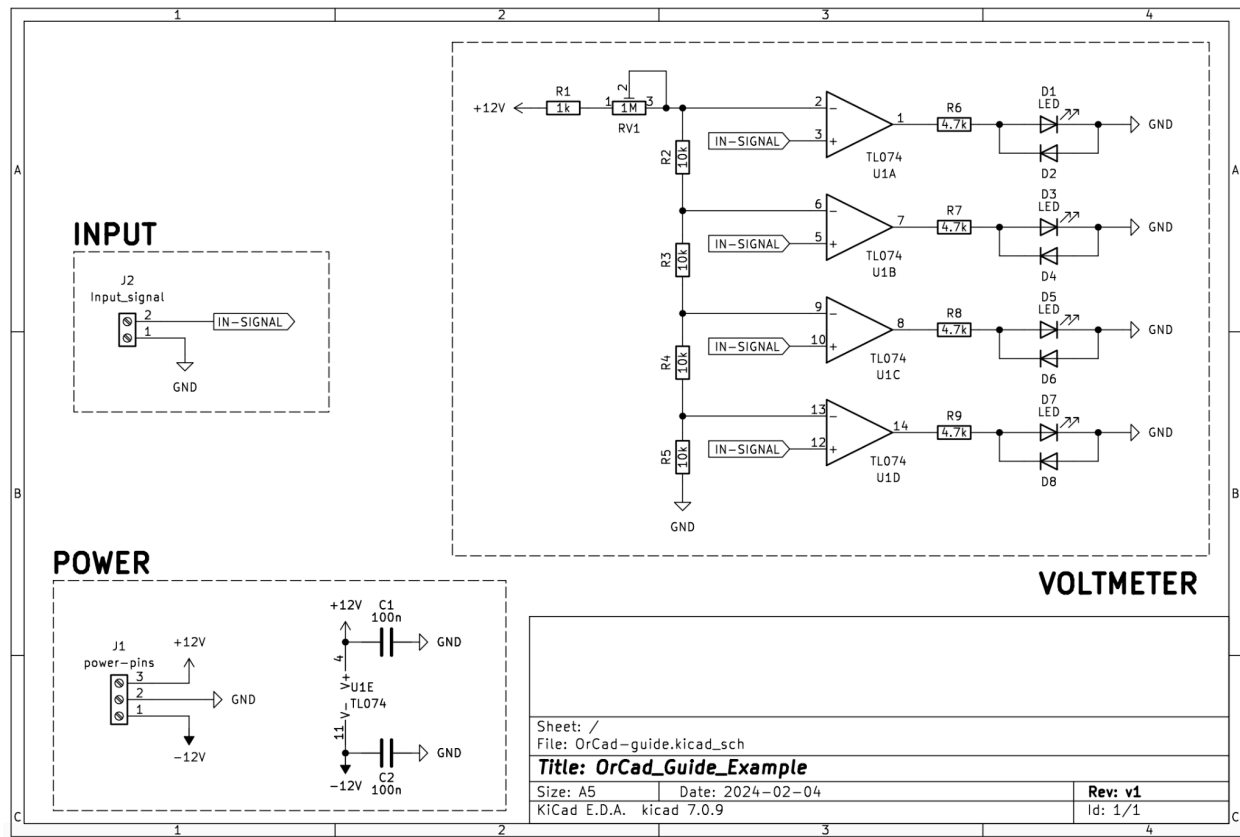
As a fair warning, this document is not official and simply exists as an aid. The best way to prevent errors is to know what you are doing, or checking against the OrCad documents.

¹ UofG, 2022, "University of Glasgow Electronic Workshop PCB Manufacture Instructions", v1.3

2 - Schematic

The schematic process is similar to OrCad, components are placed using their symbols and connections (nodes) are created to complete the circuit. KiCad has a wide range of footprints in its default library, in addition to their relative footprints for the PCB.

This guide will follow the design of a simple voltmeter as an example. The input signal is provided through connectors, and there are +12V and -12V power rails. The circuit itself is a chain of op-amps configured as comparators, comparing the input signal with tunable thresholds. The LEDs will light up when the signal goes over their comparator's threshold.



- Schematic for the simple Voltmeter.

NOTE - The circuit is very simple with little to no protection (e.g. no reverse-polarity protection on the power rails) or optimization (e.g. removing the -12V rail with a single-supply op-amp). The aim of this circuit is just to bring visual images to the guide.

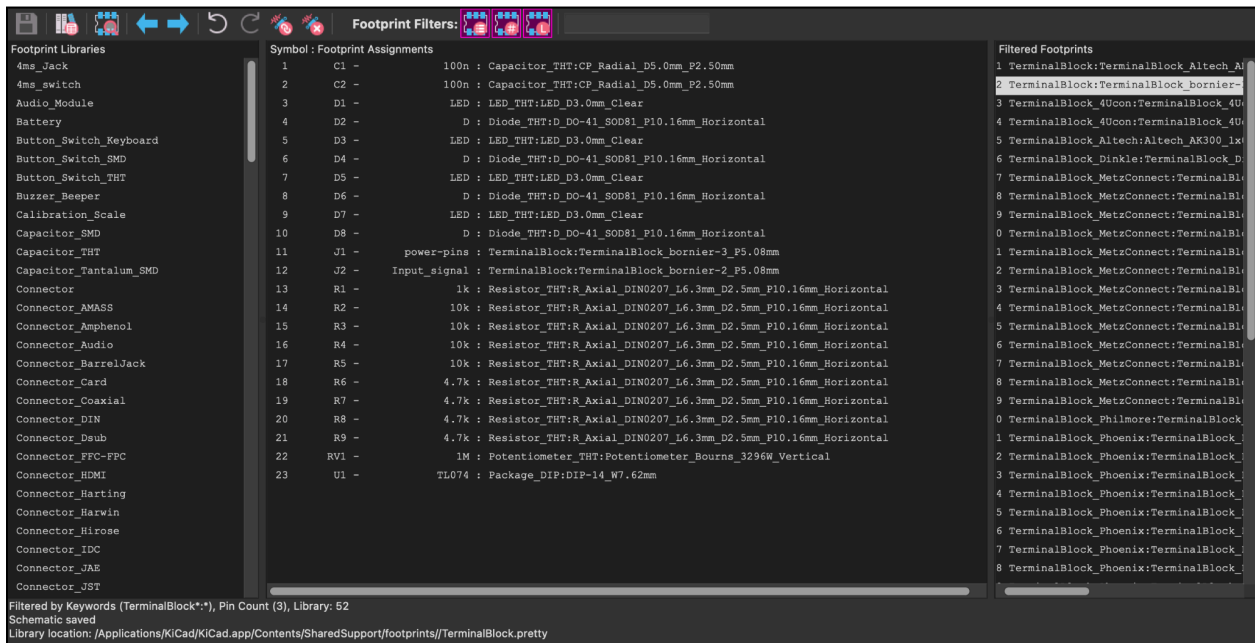
3 - Footprints

Most symbols used from the KiCad default symbol library should have at least one footprint matching from the default library. There are some subtleties as most footprints in that library assume the PCB will have a solder mask, which the UofG manufactured PCB does not.

A good way to check if the footprint won't be problematic is to look for a similar one in the "PCB editor footprint library" given by the University². If there are no similar footprints you'll have to make sure that the exposed copper tracks will not be a problem when you put the component on the board with the footprint you selected. For example, exposed metal or conductive material on the bottom of a component could short circuit to a ground plane or anything exposed on the PCB underneath.

Additionally, the recommended pad size is twice the diameter of its hole. However, make sure that pads have a diameter **bigger** than twice the diameter of their hole, for example a 1mm hole should have a pad that is at least ~2.5mm to leave enough margin for error. That might mean creating new footprints from the default ones, with the only difference being pad sizes.

NOTE - You can always remove copper from the PCB underneath the component with Rule Areas on KiCad, this will be shown later on in the guide.



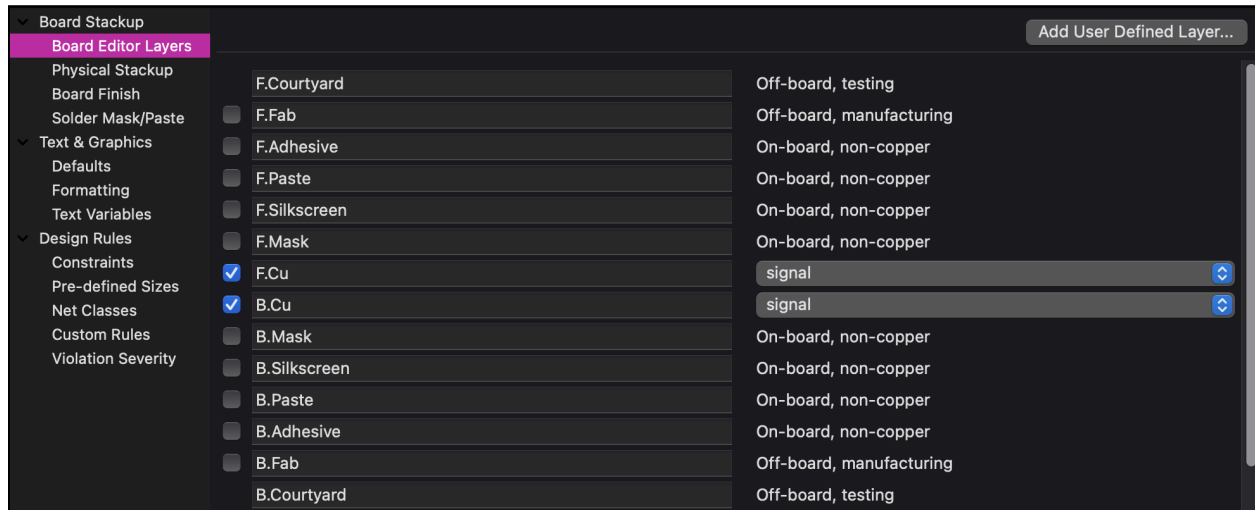
- Example footprints used for the components on the Voltmeter Schematic.

² UofG, 2014, "PCB Editor Footprints"

4 - PCB

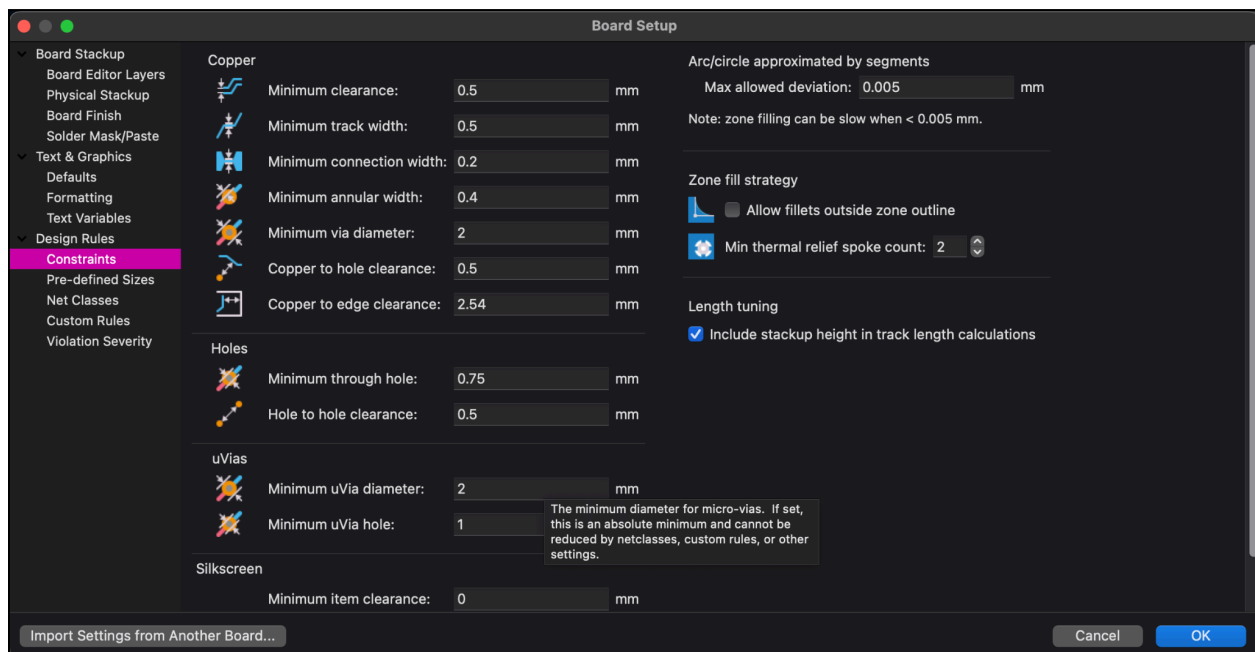
4.1 - Constraints

The UofG manufactured PCBs do not have a solder mask layer, and they also do not have a silkscreen layer (usually used for text). Hence in the “Board Editor Layers” you can remove all PCB layers except for the copper ones:



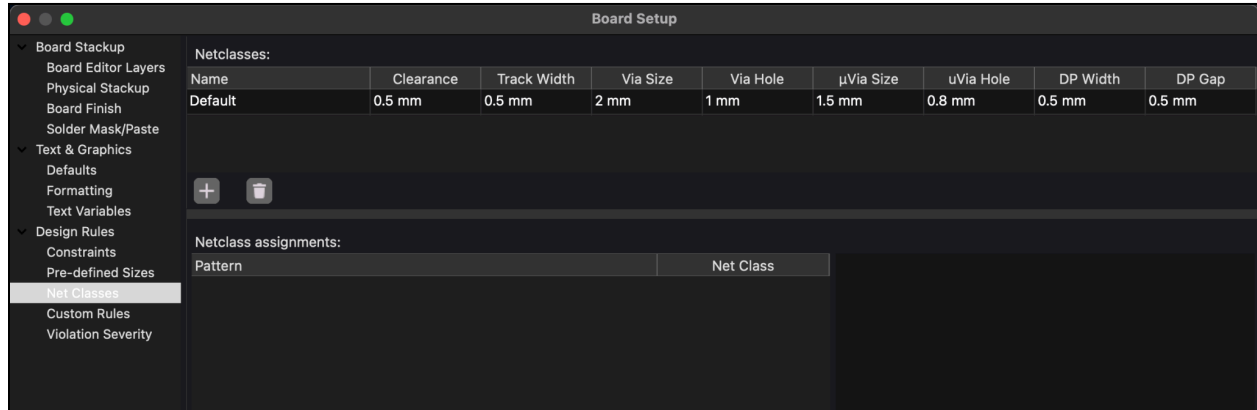
- Board Layers.

More importantly, in the Design Rules “Constraints” change the values to the values shown below. This is really important as it will either prevent designs that do not match the manufacturing capabilities, or give you an error in the DRC if you breach those capabilities.



- Design Rules Constraints.

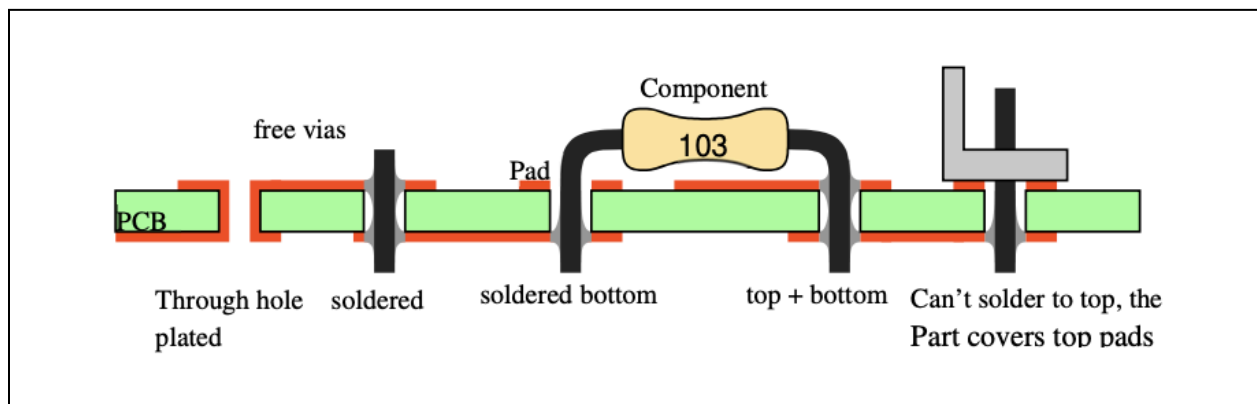
After setting the Board Editor Layers and the Design Rules Constraints, the final settings to set are the Net Classes, once again all constraints follow the PCB manufacturing guide given by the University³. Below is the minimum possible net class, but for power rails and others you can create new net classes with possibly bigger parameters (e.g. track width for current).



- Design Rules Net Classes.

From there you can create a board on the “Edge.Cuts” layer, place the components, route the components, and most importantly make sure the routing will match how you can solder the components as the UofG manufactured PCBs do not have through-hole plating.

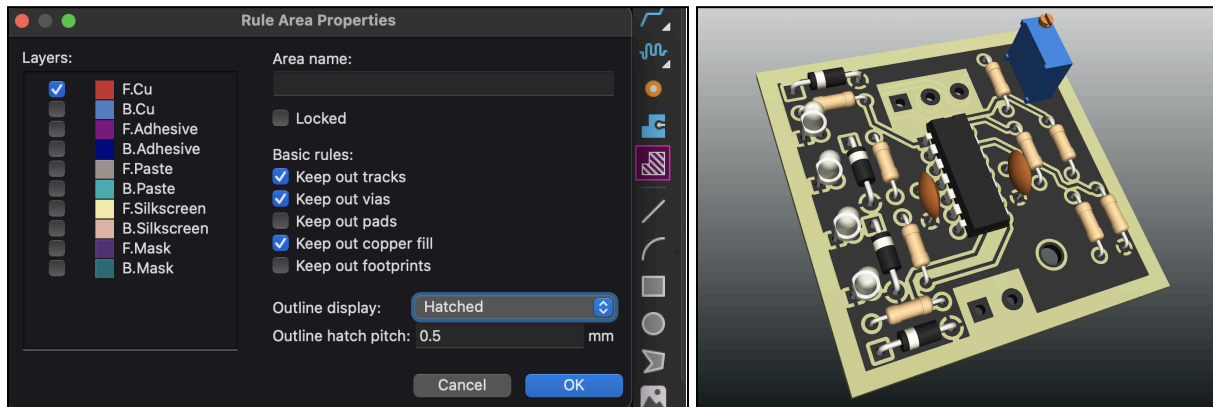
NOTE - KiCad's 3D viewer is a good way to know on which layer you can/can't solder the components. Don't forget to remove the solder mask layer in the settings of the 3D viewer to match the Uni PCBs.



- Soldering to Top/Bottom layers based on Components.

³ UofG, 2022, “University of Glasgow Electronic Workshop PCB Manufacture Instructions”, v1.3

If having a ground plane or any tracks under a component is a risk for short-circuit, KiCad has a Rule Area tool you can draw below the component that will remove any copper or trigger an error in the DRC if there is any. The setting for those and its effect are shown below.



- Rule Area (left) and the effect on the PCB with the empty rectangles (right).

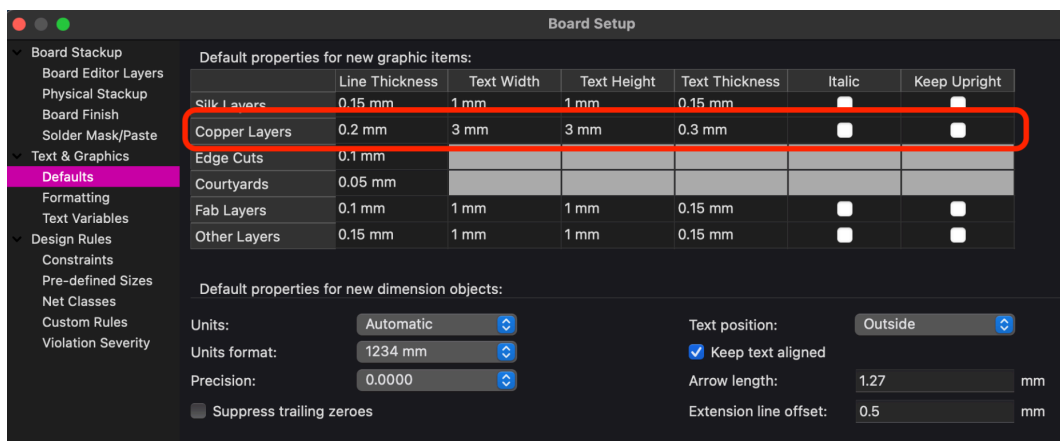
Make sure you leave space on the board edges, the image above shows how footprints for components stay within the “copper” part of the PCB (away from the edge). The mechanical holes need to be placed where there is enough material to hold the PCB, and make sure to consider the fastener’s head by leaving space around the hole (e.g. for a screw-head).

Once the design is complete **don’t forget to run the DRC**. This will ensure that the manufacturing constraints are checked before going further with the production files.

4.2 - Text / Graphic Items Constraints

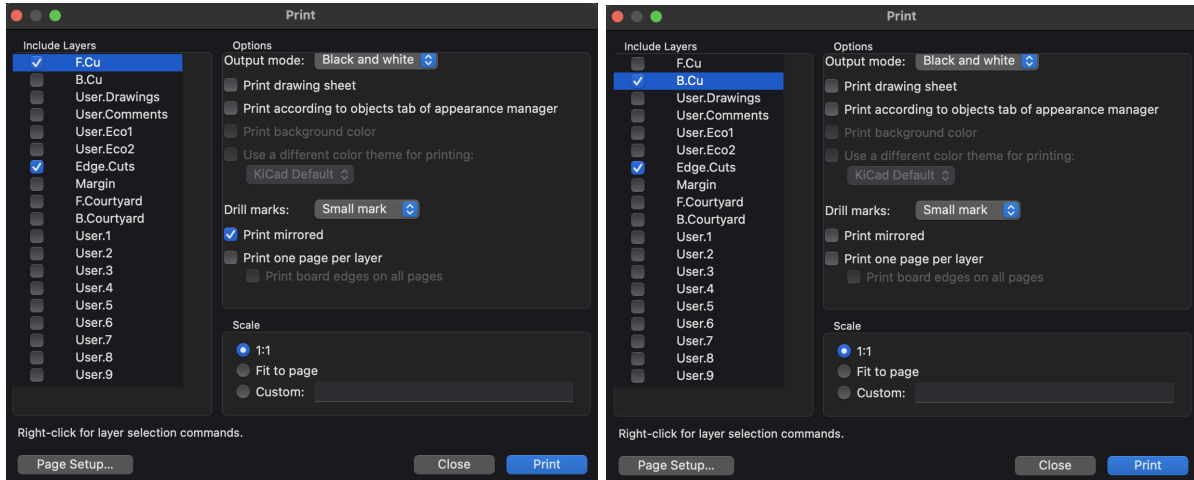
If a PCB needs text or graphic items (e.g. date, power indication, team number, etc.) some constraints will apply, those can be set in “Text & Graphics” Defaults settings. The image below should give legible text, but you can always make it bigger. Keep in mind that:

- Text has to be on copper layers, the minimum height is 3.0mm and thickness is 0.2mm
- Physical constraints also apply to text/graphics, that includes clearances, run the DRC!
- Text on the bottom layer needs to be mirrored



4.3 - Photomasks

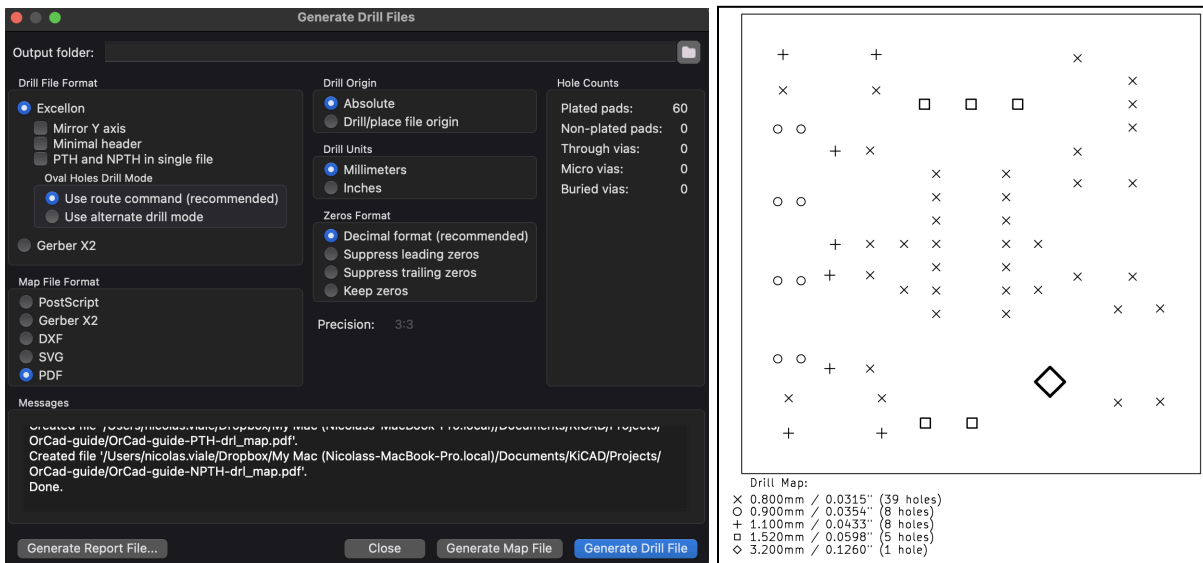
The photomask of each copper layer is essentially a black and white print of the layer with the copper in black, the edge of the PCB as a black rectangle, and the drill centres as a small white dot. KiCad has a very straight-forward print tool that can output photomasks, the settings for both layers are shown below-



- Print set-up for mirrored top (left) and bottom (right) photomasks.

4.4 - Drill Files

The drill files are created from “Fabrication Outputs” -> “Drill Files” with the following options to get both the drill files and the map file -



- Print set-up for drill and map files and Map Files.

4.5 - Submitting the Job Request

To summarise, the files you can submit for a job request are -

Top Layer Photomask (mirrored)	.pdf
Bottom Layer Photomask	.pdf
Drill Files	.drl (can be zipped to be submitted)
Map Files	.pdf

The technicians from the workshop might ask for changes/improvements from those files but this guide should be complete, going from a schematic to PCB production files ready and compatible with the UofG PCB manufacturing.

